



# India Hydrological Outlook

Based on daily observations till 31st August 2026

Issue date: 01.09.2026

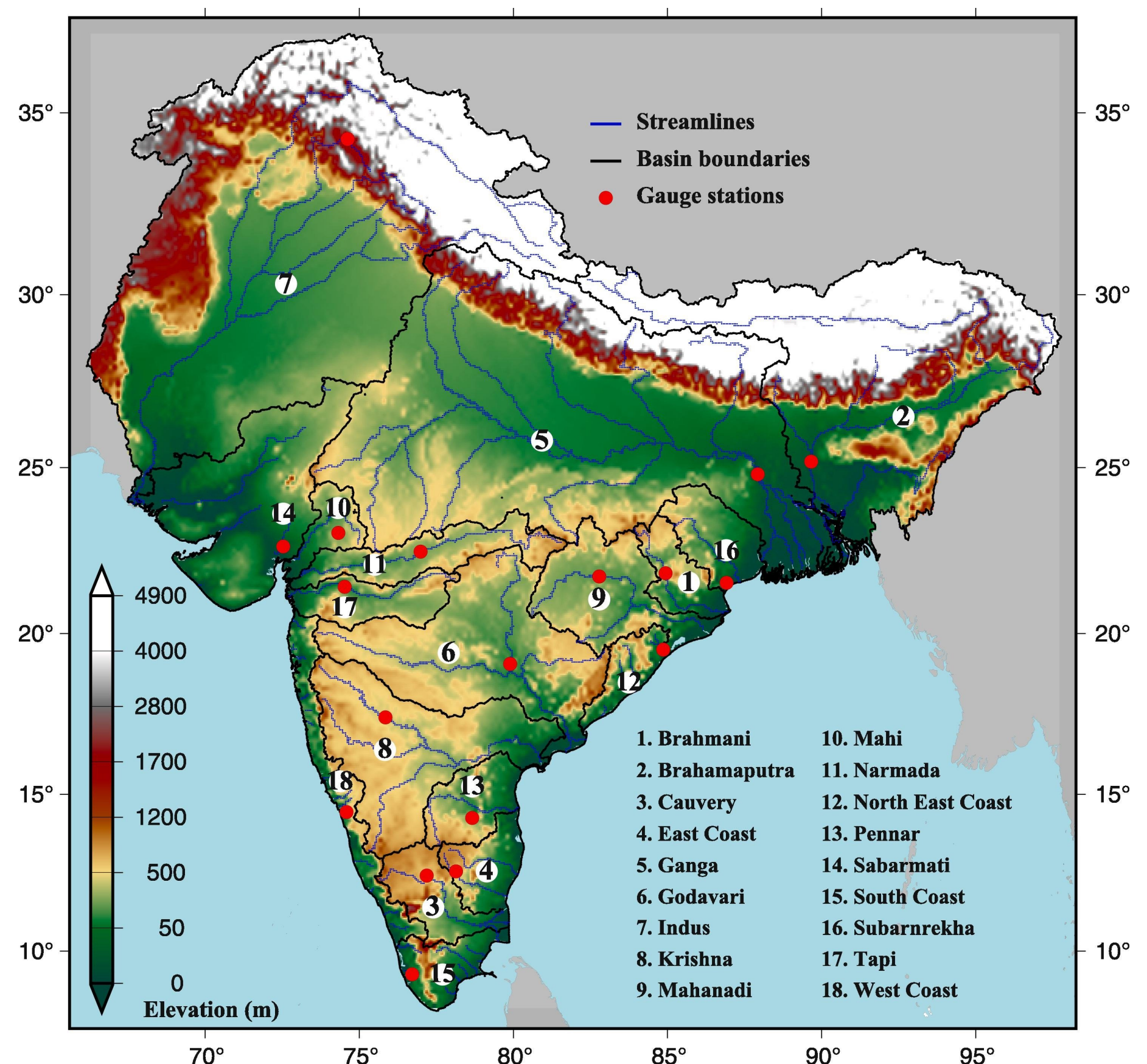
**SUMMARY:** In August, rainfall was mixed across the country, with the east experiencing higher-than-normal conditions, while the west and south recorded very low rainfall; surface air temperatures were mixed, leaning toward warmer-than-normal across most regions, and relative wetness remained near-normal across the majority of India. Total runoff showed a high deficit across the north, northeast, central, east, and south, with the west showing a surplus in August. Evapotranspiration was mixed, with elevated evapotranspiration noted in the north, northeast, and east during August. For September, rainfall is forecast to be mixed, with very high rainfall expected in the northeast and higher-than-normal rainfall in the south, while temperatures are forecast to be much cooler in the north and northeast. Total runoff is forecast to remain high in the north, northeast, central, east, and south, and evapotranspiration is forecast to be elevated across the northeast and east.

## Rainfall and Temperature:

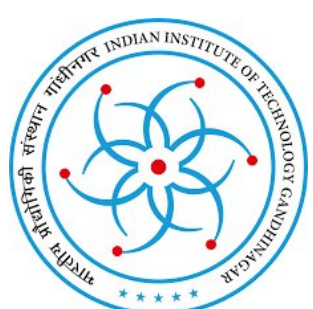
In August, rainfall across the region was mixed, with the east experiencing higher-than-normal conditions while the west saw very low rainfall. For September, the northeast is forecast to receive very high rainfall, whereas the rest of the region is expected to see mixed conditions. Surface air temperatures in August were mixed, with the south and central parts recording warmer-than-normal conditions. Temperatures are forecast to be much cooler in the northeast for September, and warmer-than-normal across the south and east.

## Soil moisture, Total runoff, and Evapotranspiration:

In August, soil moisture conditions were mixed across the region, with total runoff showing a high deficit in the north, northeast, central, east, and south, and a surplus in the west. Evapotranspiration was elevated across most areas. For September, soil moisture is forecast to be drier-than-normal in the south and northwest, while total runoff is predicted to remain in deficit across the north, northeast, northwest, central, east, and south. Evapotranspiration is forecast to be elevated in the north-east and central-east.



India Hydrological Outlook provides a comprehensive monthly summary of key meteorological and hydrological variables for current conditions alongside a four-month retrospective and one month forecast. For more information, please visit the website: [www.indiahydrolook.in](http://www.indiahydrolook.in)



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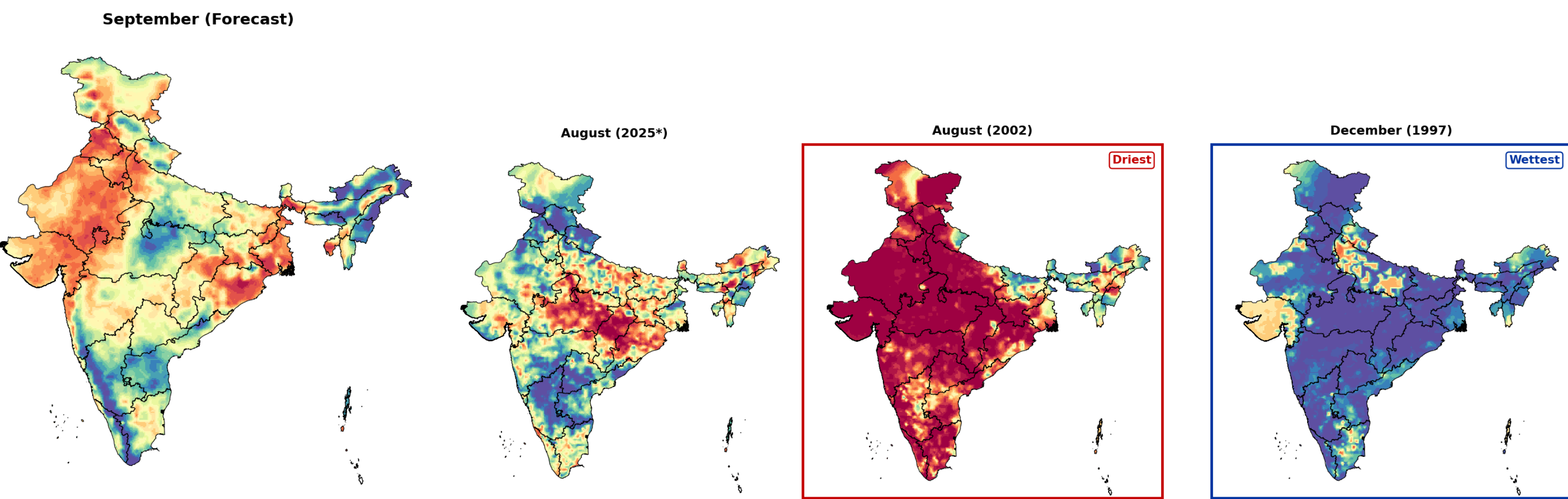
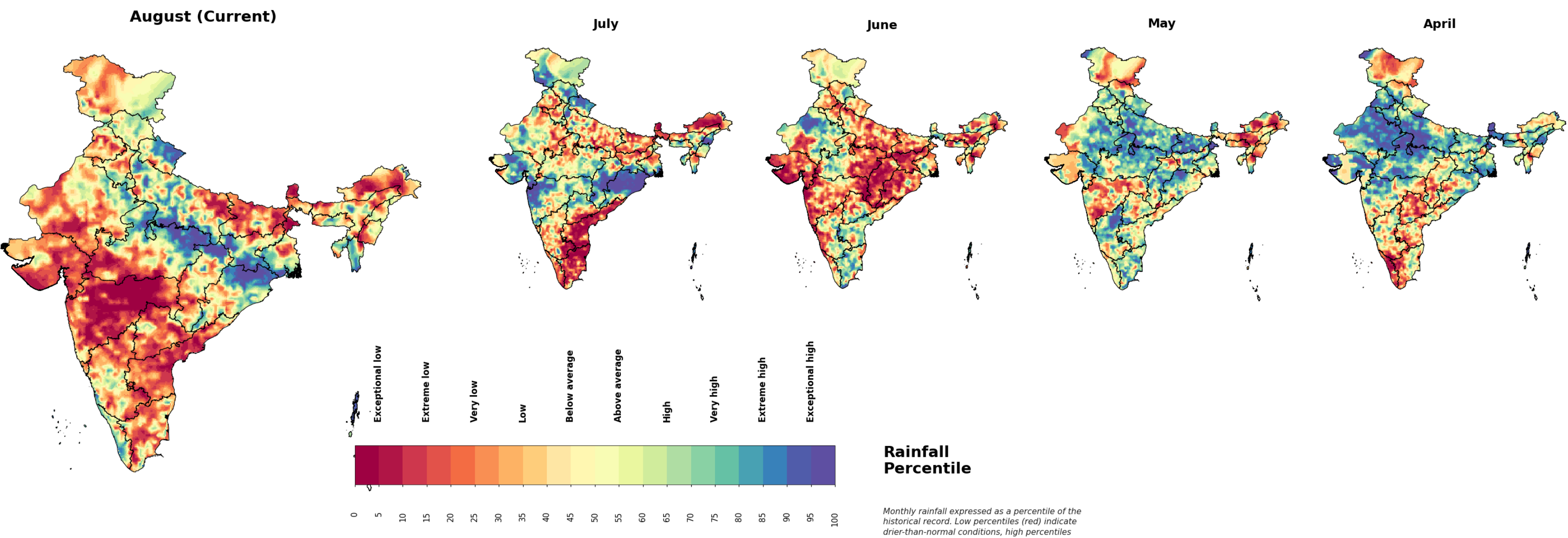
# Observed and Forecast Rainfall Conditions

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These maps are prepared from daily gridded rainfall observations and Extended Range Forecast System (ERFS) from India Meteorological Department (IMD), expressed as total monthly rainfall in percentile for period 1955 to present. These rainfall percentile maps highlight the regions with unusual dry, wet, or near-average rainfall for that time of year. High rainfall percentiles in forecast month, along with high relative wetness (maps available on page 4) in current month, may indicate a potential region for unusual wet conditions. Areas with low precipitation percentiles in forecast month, particularly those with minimal or no rainfall along with high relative dryness, may be at risk of drought conditions.

**SUMMARY:** Rainfall conditions in August showed mixed patterns across the country, with some regions experiencing higher rainfall in the east, while the northeast recorded lower-than-normal amounts.



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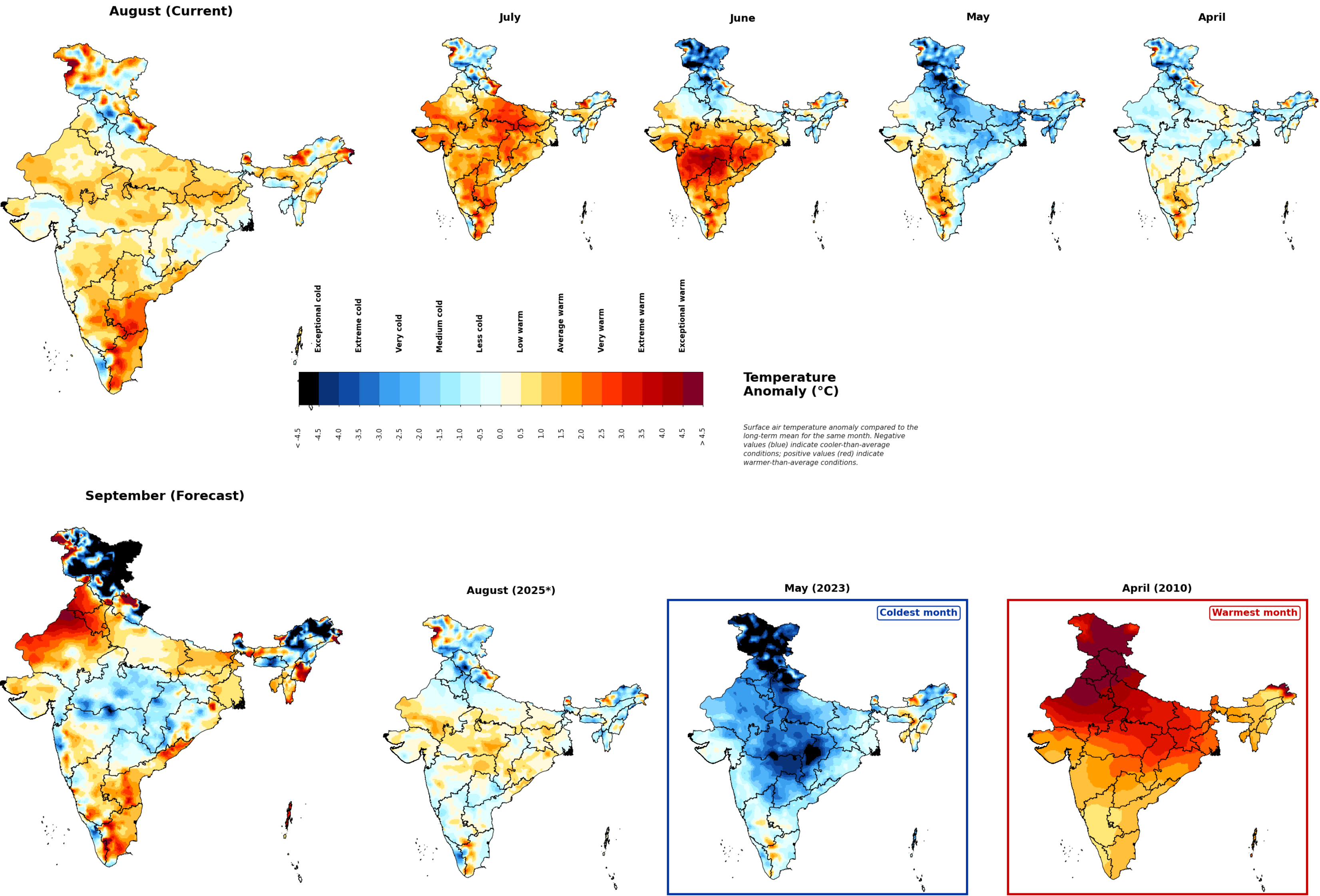
# Observed and Forecast Surface Air Temperature

Based on daily observations till 31st August 2026

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These maps are prepared from daily gridded temperature observations and Extended Range Forecast System (ERFS) from India Meteorological Department (IMD), expressed as a monthly-averaged temperature anomaly from the historical mean (1955 to present). These temperature anomaly maps highlight regions with higher and lower temperatures than usual. Low-temperature anomalies may lead to a decrease in crop yield or delayed crop growth. Areas with higher temperature anomalies paired with relatively low rainfall and low soil moisture (maps available on page 4) may be at risk of drought conditions with increased potential evapotranspiration.

**SUMMARY:** Surface air temperatures in August were warmer-than-normal across the north, northeast, northwest, central, east, and south. For September, temperatures are forecast to be much cooler than normal in the north and northeast, while the central and central-east parts are expected to be cooler-than-normal.



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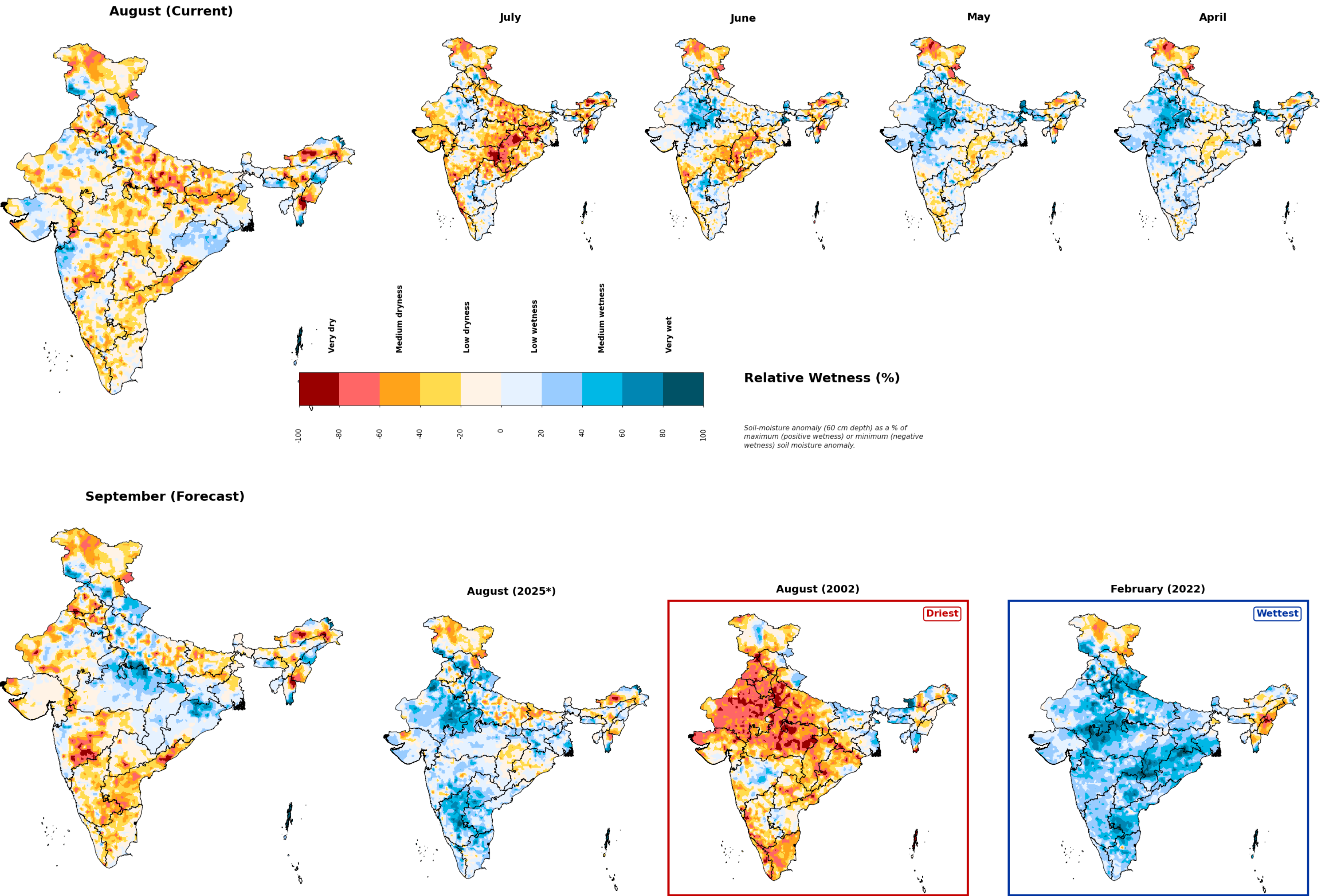
# Observed and Forecast Relative Wetness

Based on daily observations till 31st August 2026

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These maps are prepared from daily simulated soil moisture (60 cm depth) using gridded meteorological forcing from observations and Extended Range Forecast System (ERFS) from India Meteorological Department (IMD), expressed as a monthly-averaged soil moisture anomaly from the historical mean (1955 to present). The soil moisture anomalies are presented as relative to historical extremes to highlight the regions with unusual wet or dry conditions. High relative wetness (wetter) in current month paired with a high rainfall percentile in forecast month may lead to above-normal flows and unusual wetter conditions in the coming days/weeks. Areas with low relative wetness (drier) in current month and no or minimal predicted rainfall in forecast month may be at risk of lower than normal flow and moisture availability.

**SUMMARY:** Soil moisture conditions in August were mixed across the subcontinent, while the forecast for September indicates drier-than-normal conditions in the southern regions. Central and eastern parts are expected to remain near-normal.







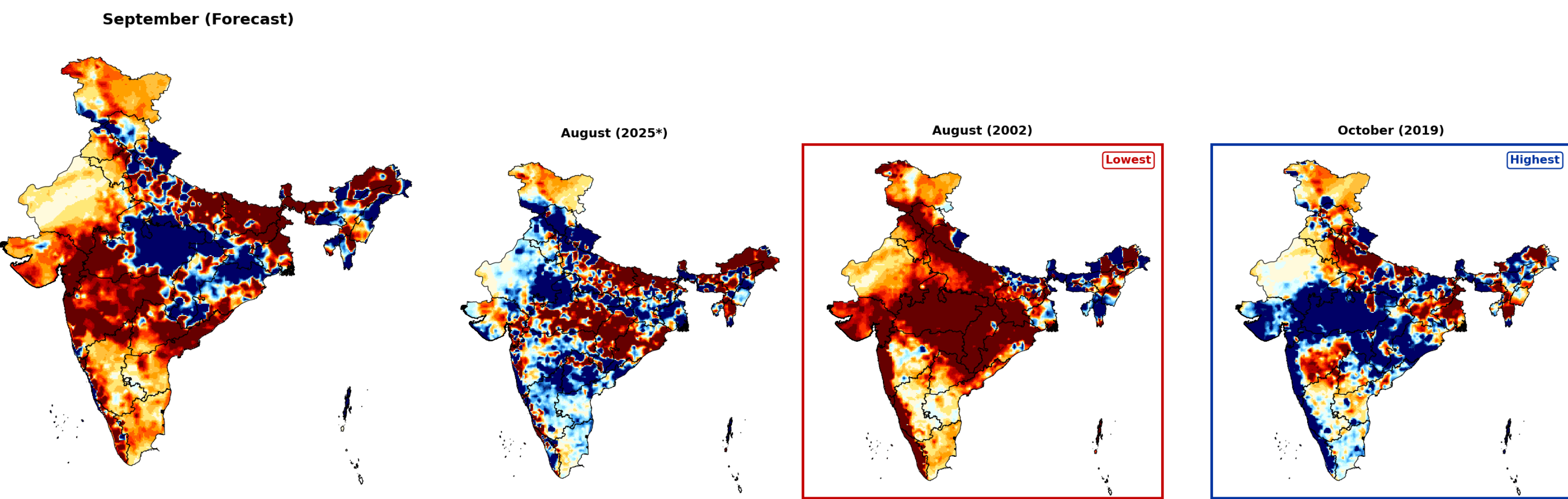
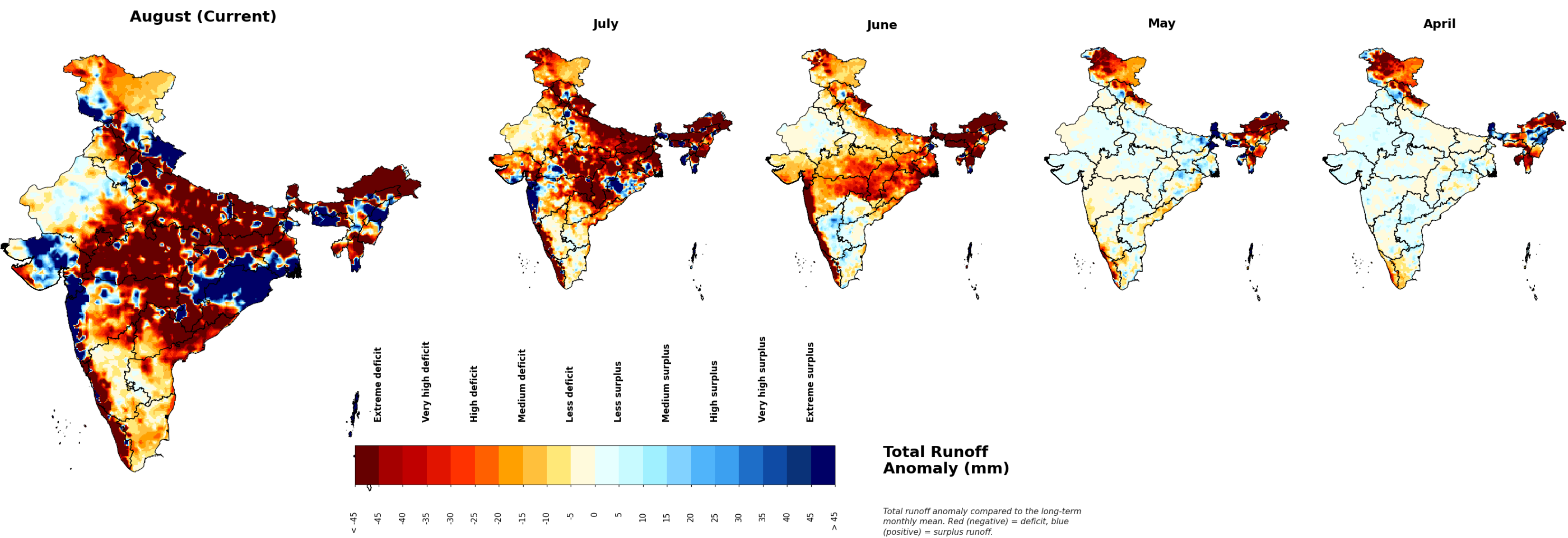
# Observed and Forecast Total Runoff

Based on daily observations till 31st August 2026

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These maps are prepared from daily simulated total runoff using gridded meteorological forcing from observations and Extended Range Forecast System (ERFS) from India Meteorological Department (IMD), expressed as a total runoff monthly anomaly from the historic mean (1955 to present). These total runoff anomalies highlight the regions with relative surplus or deficit in total runoff than typical for that time of year. A surplus in total runoff in forecast month paired with relatively higher soil moisture in current month may lead to above-normal streamflow in the coming days/weeks. Areas with a high deficit in total runoff in forecast month, particularly those with high dryness in current month, may be at risk of relatively low streamflow in the coming days/weeks.

**SUMMARY:** Total runoff exhibited a high deficit across the north, northeast, central, and east regions in August, while the west showed a surplus.







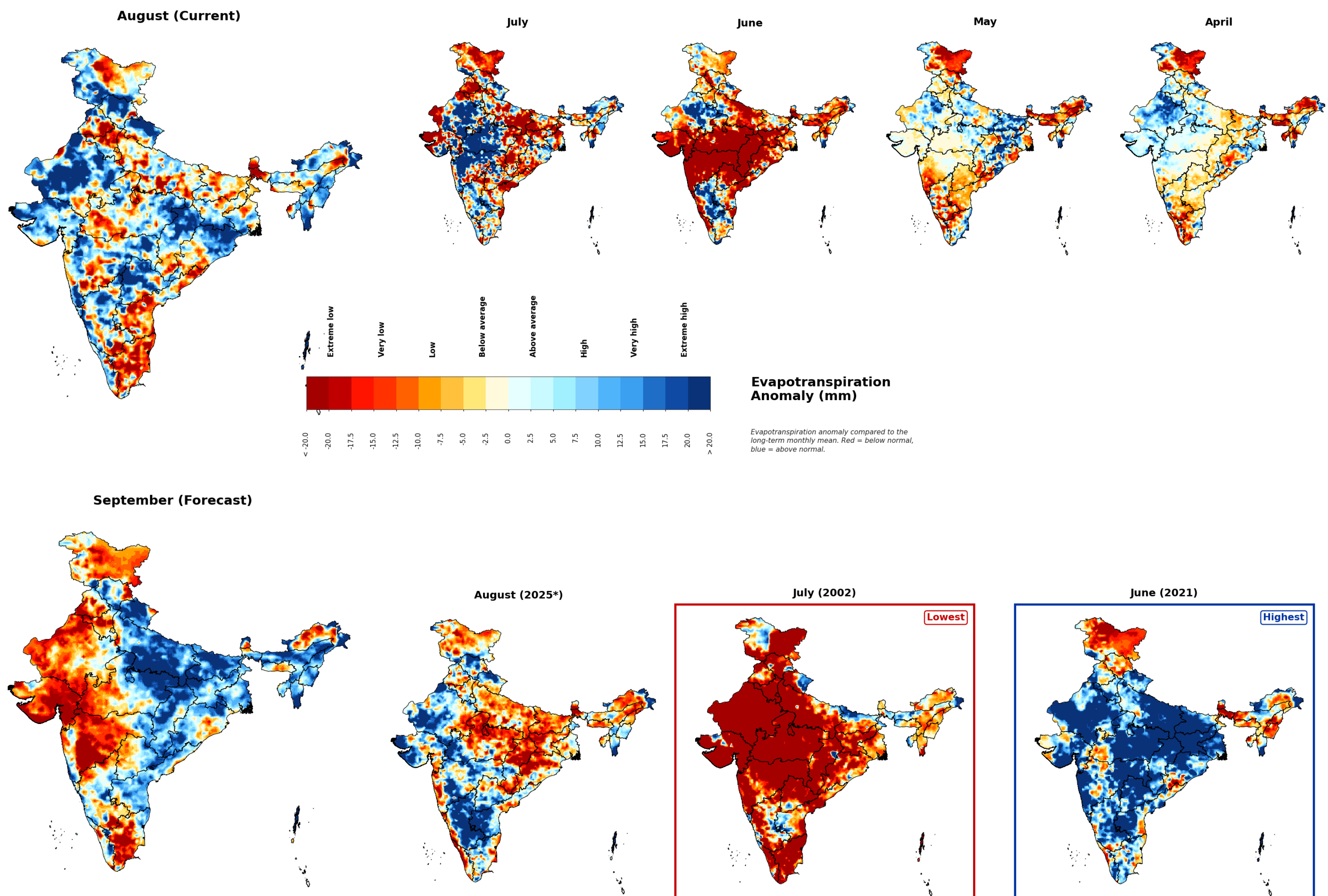
# Observed and Forecast Evapotranspiration

Based on daily observations till 31st August 2026

Issue date: 01.09.2026

These maps are prepared from daily simulated evapotranspiration (ET) using gridded meteorological forcing from observations and Extended Range Forecast System (ERFS) from India Meteorological Department (IMD), expressed as a total monthly ET anomaly from the historic (1955 to present). These ET anomalies highlight the regions with less or more water lost than usual from land and plants, helping to pinpoint areas at risk of extreme weather impacts, especially crop water stress. Low ET anomalies indicate increased crop water stress and relatively less moisture availability. High ET anomalies may lead to rapid depletion of soil moisture (flash drought) or may worsen the pre-existed droughts.

**SUMMARY:** In August, evapotranspiration conditions were mixed across the country, leaning toward elevated rates in the north, northeast, and central parts. For September, the forecast indicates mixed conditions, with elevated evapotranspiration expected in the northeast and central-east.



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# India Hydrological Outlook

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## About the India Hydrological Outlook

This document provides a comprehensive water outlook for India, covering the current month, the past four months, and a one-month forecast. The outlook is generated using observational data and an extended-range forecasting system for meteorological variables, combined with advanced hydrological modeling tools. These tools analyze relative changes and predict water availability across various regions, offering valuable insights into hydrological situation in India. Developed by the Water and Climate Lab at IIT Gandhinagar, this outlook aims to support water resource management, planning, and decision-making by offering reliable information on rainfall, temperature, soil moisture (60 cm), runoff, and streamflow patterns across different timescales.

## Datasets

The India Hydrological Outlook is made possible through the valuable cooperation of numerous data providers, whose contributions are gratefully acknowledged. Contemporary daily observations and extended range forecast data are supplied by the India Meteorological Department (IMD) [https://www.imdpune.gov.in/cmpg/Griddata/Rainfall\\_25\\_Bin.html](https://www.imdpune.gov.in/cmpg/Griddata/Rainfall_25_Bin.html). These datasets are used for initializing hydrological models and generating reliable outlook information through statistical analysis of historical analogues. Hydrological model is calibrated and validated using daily streamflow for sufficient long duration from India-Water Information System <https://indiawris.gov.in/wris#RiverMonitoring>.

## Hydrological Model

The India Hydrological Outlook utilizes the Variable Infiltration Capacity (VIC) model, a large-scale, semi-distributed hydrological model that computes water and energy budgets for each grid. The VIC model captures sub-grid variability in vegetation and elevation, providing a more accurate representation of hydrological processes at a regional scale. VIC model simulations are performed at a daily temporal resolution to provide accurate hydrological forecasts and enhance water resource management across India.

## Disclaimer and Liability

The India Hydrological Outlook aims to ensure that all content provided is accurate and aligns with the current scientific understanding. However, the science underlying the meteorological and hydrological forecasts, as well as climate projections, is continuously evolving. As such, any forecast or prediction included in the content should not be regarded as a definitive statement of fact. To the fullest extent permitted by applicable law, the India Hydrological Outlook disclaims all warranties or representations, whether express or implied, regarding the content. Your use of the content is entirely at your own risk, and we make no guarantees that the content is error-free or suitable for your specific needs.

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India Hydrological Outlook Portal: [www.indiahydrolook.in](http://www.indiahydrolook.in)



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